

ISyE 6669

Project Report

Parisa Babolhavaeji, Mahsa Hedayati, Vedika Sanjeev Lakhanpal

1 Electrolyser

Assume that the electrolyser is added but no other changes are made.

- (a) How does the LP change to include the electrolyser in operation?

New Decision Variable

$y_e[i] \in \{0, 1\}$: where the electrolyser is ON at hour i

LP with Electrolyser:

$$\min \sum_{i=1}^N c_e[i] \cdot x_e[i] + 10 \cdot x_h[i]$$

Subject to:

$$x_e[i] + s[i] \geq r_e[i] + 0.5 \cdot y_e[i] \quad \forall i \in [1 \dots N] \quad [\text{Electricity requirements}]$$

$$x_h[i] + \sum_{k=1}^{i-1} (x_h[k] + 9 \cdot y_e[k] - r_h[k]) \geq r_h[i] \quad \forall i \in [1 \dots N] \quad [\text{Hydrogen demand + storage}]$$

$$x_e[i], x_h[i] \geq 0, \quad y_e[i] \in \{0, 1\}$$

- (b) Implement your new model in **Python**.

Total operational cost with electrolyser: \$38,546.27

- (c) Comment on how your solution and costs have changed.

With the electrolyser added, the total operational cost dropped from \$44,409.00 (baseline) to \$38,546.27. The model chose to operate the electrolyser consistently, which shows that producing hydrogen on-site was generally more cost-effective than purchasing it at \$10/kg — even when accounting for electricity usage.

2 Solar

Now assume that **instead** the company decides to purchase additional solar.

- (a) Alter the original LP assuming two additional blocks of solar are purchased. How does this change the optimal operational cost?

We modified our original LP to purchase two extra 0.2 MW solar blocks, bringing the total solar capacity from 2.0 MW to 2.4 MW. The solar capacity is increased, and the decision variables for grid electricity and hydrogen purchases are defined to meet the plant's hourly requirements at minimum cost. For a 72 hour period optimal total operational cost of **\$44,342.06** was seen.

```
#additional 2 solar blocks
solar_capacity = 2.0 + 0.4 # 2.4 MW total
solar_generation = solar_forecast * solar_capacity
```

- (b) Now alter the problem to determine the optimal amount of new solar to purchase. You need to include the cost of the panels, for this purpose you can assume that the costs will be evenly distributed over 3.5 years (ignoring inflation/interest).

For this, the original LP was modified by introducing an integer decision variable S that represents the number of extra 0.2 MW solar blocks to purchase. The total solar capacity is

$$2.0 + 0.2 S \quad (\text{MW}),$$

and the hourly solar generation is obtained by multiplying this capacity by the solar forecast. Each extra block costs \$900,000, and this capital cost is amortized over 3.5 years (approximately 30,660 hours). For our 72 hour time period, the fraction of the cost incurred is

$$\frac{72}{30660}.$$

The solver returned:

```
Solver status: optimal
Optimal operational cost over 72 hours: $44,409.00
Optimal number of extra 0.2 MW solar blocks (S): 0
```

This indicates that under the given data and assumptions, purchasing additional solar capacity is not cost-effective over the 72-hour period because the amortized cost of new solar blocks exceeds the potential savings in grid electricity cost.

- (c) Implement your new problem in **Python** and determine the optimal solar installations and new **operational** cost over the three days.

- **Decision Variables:**

- $E[t]$: Grid electricity purchased in hour t (MWh).
- $H[t]$: Hydrogen purchased in hour t (kg).
- S : Number of extra 0.2 MW solar blocks (integer).

- **Constraints:**

$$E[t] + (2.0 + 0.2 S) \cdot \text{solar_forecast}[t] \geq e_demand[t], \quad \forall t,$$

$$H[t] \geq h_demand[t], \quad \forall t.$$

- **Objective:** Minimize the total cost

$$\min (e_prices \cdot E) + 10 \sum_t H[t] + \left(\frac{72}{30660} \times 900,000 \right) S.$$

After solving the problem it is clear that under the given data and assumptions, it is not cost-effective to purchase any additional solar capacity over the three day period. The amortized cost of additional solar exceeds the savings from reduced grid electricity purchases.

3 Battery

Finally **instead** consider the purchase of the battery.

1. How does the LP change to include the battery in operation? Assume the battery starts with no stored energy.

New Decision Variables

- $b_i \in [0, 0.5]$: Energy stored in the battery at the end of hour i (MWh).
- $d_i \geq 0$: Energy discharged from the battery in hour i (MWh).
- $c_i \geq 0$: Energy charged into the battery in hour i (MWh).

Modified Constraints

- (a) **Battery Dynamics:**

$$b_i = b_{i-1} + c_i - d_i \quad \forall i \in [1, \dots, N], \quad b_0 = 0$$

- (b) **Charge/Discharge Limits:**

$$0 \leq c_i \leq 0.5, \quad 0 \leq d_i \leq 0.5, \quad b_i \leq 0.5 \quad \forall i \in [1, \dots, N]$$

- (c) **Electricity Demand Constraint:**

$$x_e[i] + s[i] + d_i \geq r_e[i] + c_i \quad \forall i \in [1, \dots, N]$$

Objective Function (Unchanged):

$$\min \sum_{i=1}^N (c_e[i] \cdot x_e[i] + 10 \cdot x_h[i])$$

2. Implement the altered LP in **Python**.

Optimal operational expenditure: \$44271.00

3. Comment on how your solution and costs have changed.

The addition of a 0.5 MWh battery reduced operational costs marginally, from \$44,409.00 to \$44,271.00 (saving \$138 over 72 hours), with annual savings of \$16,790. In contrast, the electrolyser drastically lowered costs to \$38,546.27 (annual savings: \$713,298) by substituting costly hydrogen purchases (\$10/kg) with cheaper on-site production when electricity prices were below \$180/MWh.

4

The company wants to make the investment which can be recovered as quickly as possible. Calculate the buy-back times for each technology, and make a recommendation to your employer. The costs of the electrolyser and battery are shown below.

Component	Cost
0.5 MW Electrolyser	\$500,000
0.5 MWh Battery	\$75,000

Technology	Cost (\$)	Payback Time (Years)	Preferred
Battery	43791.73	1	
Solar	44,377.49	1105.1	
Electrolyser	38546.27	0.70	X

Task allocation:

Parisa- Q3, Q4

Vedika- Q2, compiling codes

Mahsa- Q1